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SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I ASWATHA · V – 192525390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

SET 1

1) CLOUD ARCHITECTURE FOR VIDEO PLATFORM

Introduction:

Cloud computing helps video streaming platforms handle a large no. of users. It provides better speed, scalability, and reliable service.

Cloud Architecture:

* load balancer: distributes among servers.
* CDN: Delivers videos from nearby server.
* cloud storage
* caching
* Auto-scaling
* Database.

Scalability:

* Add servers when users increases.
* Reduce servers during low traffic
* Auto-scaling reduces manual work
* Prevents server overload
* Supports millions of user.

Caching:

* stores frequently accessed videos
* Reduces server workload.
* Improves loading speed.
* Reduces buffering.

Benefits:

* Smooth streaming
* Less buffering
* High availability
* Better performance
* Lower cost.

Conclusion:

Cloud architecture provides fast and reliable video streaming. CDN, caching, load balancing and auto-scaling make the platform efficient and scalable.

2) MULTI-CLOUD CONTAINER DEPLOYMENT

Introduction:

A fintech startup needs a secure & reliable system to run its applications. Multi-cloud deployment allows applications to use more than one cloud provider.

Proposed solution:

* Docker: Packages applications into containers.
* Kubernetes: Manages & controls containers.
* Multi-Cloud: Runs application across different cloud providers.
* CI/CD: Automates testing & deployment
* Terraform: Helps & manages cloud infrastructure.

Service Discovery:

* Helps containers find each other.
* Uses service names instead of changing IP address.
* Communication between services easier.
* Improves application managment.

Scaling:

* Add containers & Removes containers.
* Kubernetes can manage scaling automatically.
* Helps maintain application performance.

Advantages:

* Reduces dependency on one cloud provider.
* Provides better availability
* Flexible deployment.
* Improves reliability
* Makes scaling easier.

Limitations:

* Multi-cloud management is complex.
* Monitoring costs can be difficult.
* security needs carful managment.
* Network communications may cause delays.

3. CLOUD BACKUP AND DISASTER RECOVERY

Introduction:

An E-learning platform stores important student & course data. cloud backup & disaster recovery help protect this information from accidental deletion or system failure.

Backup solution:

* Automated Backup: Creates backup regularly.
* stores backup data securely
* keeps older versions of files.
* Maintains copies in another locations.
* Protects backup data.

RPO: Recovery Point objective:

* Defines the max acceptable data loss.
* 15 mins RPO means data can be recovered close to the last 15 minutes.
* Helps decide backup frequency.

RTO: Recovery Time Objective:

* Defines how quickly the system should be restored.
* 1hrs RTO means the system should recover within 1hour.
* Helps reduce service downtime.

Benefits:

* Prevents data loss
* Reduce downtime
* Provide reliable recovery
* Protects important data & supports.

Recovery steps:

* Detects the failure
* Identify the backup & restore data
* Restart important services
* check that the system is working properly.

H) HYBRID CLOUD ARCHITECTURE

Introduction:

Government agencies handle both sensitive & non-sensitive information. A hybrid cloud allows sensitive data to remain in private infrastructure.

Proposed hybrid cloud Architecture:

Private cloud: stores sensitive government data.

Public cloud: Handles non-sensitive applications.

VPN: Secure communication

Firewall: controls Network traffic

Data classification: Decides where information should be stored or not.

Encryption :

* Encrypt sensitive data at rest.
* Encrypt data during transmission
* Protect encryption keys carefully
* change keys regulary when required.

Advantages :

* More difficult to manage.
* Requires stores security controls.
* Better security
* Flexibility in resource usage.
* Keeps sensitive data under control.
* Helps meet compliance requirements.

Limitations :

* Data transfer may cause delay & difficult to manage
* management cost may increase
* Requires strong security controls.

5) MULTI-REGION DEPLOYMENT FOR SAAS :

Introduction :

SAAS application should remain available even when a cloud region fails. Mult-region deployment provides high availability by running applications in different regions.

Proposed Architecture:

* Deploy applications in multiple cloud regions.
* Use LOAD BALANCING - to distribute traffic.
* Use DATA REPLICATION b/w regions.
* Maintain similar applications environment.
* Uses healt checks to monitor regions.

Data Replication:

* Keep copies of information in different region.
* Replication automatic
* Maintain separate backups
* encrypt repeicated data
* Regularly check data consistency.

Monitoring & Alerting:

* Monitor CPU & memory usage
* Track application errors
* Monitor network performance
* Alerts for failures.

Benefits:

* High availability
* Reduced downtime
* Better fault tolerance
* Improved user experience
* Protection from regional failures.